PatchDETR

Revised Mathematical Formulation

Version 1.0 — 27 October 2026

Integrating PatchCvT-style memory evidence with DETR for degraded traffic channelizing devices

Purpose. This document proposes a cleaner mathematical integration based on Autopilot's conceptual review. The original PatchCvT formulation remains a reference; proposed PatchDETR changes are explicitly identified.

Status. Proposed formulation, not an empirically validated model. Existing data support an exploratory proof of concept (POC), not claims of detection performance or generalisation.

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1 Executive summary

The current prototype mixes frozen-feature memory retrieval, residual energy, a learned robustness score, a multiplicative gate, a centre heatmap and DETR losses. The review identified mismatches between what some terms are named to represent and what they mathematically optimise.

- **Keep:** a frozen pretrained backbone; memory retrieval as input-side evidence; DETR set prediction; paired clean/degraded training as an optional invariance objective.
- **Remove from the default formulation:** $\mathcal{L}_{\text{mem}}$, $\mathcal{L}_{\text{rob}}$ targeting $\exp(-E)$, $\mathcal{L}_{\text{gate}}$ and the compensating $\mathcal{L}_{\text{gate_preserve}}$.
- **Change:** separate object/background banks and pass their evidence as distinct features; call the existing centre heatmap *objectness*, not uncertainty.
- **Defer:** attention priors, predicted box uncertainty, top-k query seeding and a multi-scale neck until the basic evidence signal is validated.

Central hypothesis. Memory evidence may help DETR preserve useful visual cues under degradation. Whether it distinguishes damaged TCDs from background remains unproven.

2 Findings from the current formulation

The single-bank residual energy is

$$E_j = \|q_j - \widetilde{q}_j\|_2.$$

Reported values were approximately 0.740 on clean TCD tokens, 0.759 on mud-degraded TCD tokens and 0.711 on background. These values are close and do not support the claim that high E means a damaged TCD. A mixed object/background bank measures closeness to some familiar patch, not specifically closeness to an intact object.

- With frozen backbone and fixed memory, $\mathcal{L}_{\text{mem}} = \text{mean}(E)$ has no trainable gradient path.
- Regressing R toward $\exp(-E)$ is self-distillation of a deterministic input-derived quantity, not an independent robustness target.
- A gate S that decreases as E rises can suppress degraded tokens, opposing the intended preservation goal.
- The current CenterNet-style heatmap is supervised with object-centre targets; it estimates objectness, not uncertainty.
- The motion prior was higher inside objects in the observed data, but camera ego-motion may confound this signal.

These findings justify simplifying the formulation. They do not prove that PatchCvT as a broader concept is invalid.

3 Design constraints and notation

- The pretrained backbone remains frozen: unfreezing caused rapid collapse in previous experiments.
- The memory banks remain fixed during detector training.
- Object boxes are small; stride-8 features or a validated multi-scale representation are worth considering.

- No comprehensive damaged-TCD labels exist, so damage-specific conclusions require manually tagged examples.

Symbol	Meaning
x, x^+	Original/clean image and its paired degraded version.
f_θ, F, q_j	Frozen backbone, spatial features and L_2-normalised token.
$M_{\text{obj}}, M_{\text{bg}}$	Fixed object and background memory banks.
$d_{j,\text{obj}}, d_{j,\text{bg}}$	Token distance to each bank.
e_j	Evidence vector supplied to the detector.
$\mathcal{L}_{\text{det}}$	Standard DETR set-prediction loss.

4 Proposed PatchDETR formulation

4.1 Frozen visual features

$$F = f_\theta(x) \in \mathbb{R}^{H\times W\times C}, \qquad \theta \text{ frozen}, \qquad q_j = \frac{F_j}{\|F_j\|_2 + \varepsilon}. \tag{N1}$$

No gradient flows into f_θ. A trainable projection or neck downstream is allowed without unfreezing the backbone. Stride-8 features are recommended for the small objects, subject to validation.

4.2 Separate object and background banks

$$M_{\text{obj}} = \{m^k_{\text{obj}}\}_{k=1}^{K_{\text{obj}}}, \qquad M_{\text{bg}} = \{m^k_{\text{bg}}\}_{k=1}^{K_{\text{bg}}}. \tag{N2}$$

Build banks from labelled memory images: tokens inside ground-truth TCD boxes enter M_{obj}; sampled tokens outside boxes enter M_{bg}. Keep the banks fixed. This separation makes the comparison object-specific rather than generic novelty-only.

4.3 Distance-to-bank evidence

$$d_{j,s} = -\tau \log\left[\frac{1}{K_s}\sum_{k=1}^{K_s}\exp\left(-\frac{\|q_j - m^k_s\|_2}{\tau}\right)\right], \qquad s \in \{\text{obj}, \text{bg}\}. \tag{N3}$$

The $1/K_s$ normalisation avoids a bank-size-dependent additive offset. For unit-normalised features, Euclidean distances lie in $[0, 2]$; this soft-min approximates nearest-bank distance, with temperature effects. A k-nearest-neighbour mean is a simpler alternative. Distances have no training gradient when features and banks are fixed.

$$e_j = [d_{j,\text{obj}}, d_{j,\text{bg}}, m_j], \qquad m_j = d_{j,\text{bg}} - d_{j,\text{obj}}. \tag{N4}$$

- Positive m_j means the token is closer to the object bank than to the background bank.
- A large $\min(d_{j,\text{obj}}, d_{j,\text{bg}})$ means the token resembles neither bank particularly well.
- Neither distance nor margin is automatically an anomaly label or uncertainty estimate.

The two-bank margin is a hypothesis to validate, not a demonstrated solution.

4.4 Evidence-conditioned tokens

$$u_j = W\left[q_j \,\|\, (\widetilde{q}_{j,\mathrm{obj}} - q_j) \,\|\, e_j\right] + p_j, \qquad T = \mathrm{Encoder}\left(\{u_j\}_j\right). \tag{N5}$$

Here $\widetilde{q}_{j,\mathrm{obj}}$ is the soft-retrieved feature from the object bank, p_j is positional encoding, and W is a trainable projection. Feed these tokens to the DETR decoder. Do not multiply them by an anomaly-dependent gate in the minimum formulation.

Minimal alternative. First visualise the evidence maps or use them to re-score detections from a baseline detector. This can test whether the memory signal is useful before training the full integration.

5 Detection and invariance losses

$$\mathcal{L}_{\mathrm{det}} = \mathcal{L}_{\mathrm{cls}} + \lambda_{\mathrm{box}}\mathcal{L}_{L_1} + \lambda_{\mathrm{giou}}\mathcal{L}_{\mathrm{GIoU}}. \tag{N6}$$

Use standard Hungarian matching. The existing coefficients $\lambda_{\mathrm{box}} = 5$, $\lambda_{\mathrm{giou}} = 2$ and no-object weight 0.1 may be retained as engineering defaults, not theoretical optima.

$$\mathcal{L}_{\mathrm{pair-det}} = \frac{1}{2}\left[\mathcal{L}_{\mathrm{det}}(x) + \mathcal{L}_{\mathrm{det}}(x^+)\right]. \tag{N7}$$

Detection supervision on both original and degraded images is the primary robustness driver.

$$\mathcal{L}_{\mathrm{inv}} = \frac{1}{|A|}\sum_{j\in A}\left[1 - \cos\left(t_j(x^+), \mathrm{stopgrad}(t_j(x))\right)\right]. \tag{N8, optional}$$

A contains spatially aligned tokens or matched object regions. The clean branch is a detached target; gradients flow through the degraded branch into trainable projection/encoder layers. Backbone and banks remain fixed. Alignment must account for geometric augmentation. Stop-gradient alone does not guarantee against collapse, so detection supervision must remain dominant. Start with $\lambda_{\mathrm{inv}} = 0$, then test a small value such as 0.1 as an ablation.

6 Attention prior (optional, not in the minimum POC)

$$\pi_j = \sigma\left(w_0 + w_1 d_{j,\mathrm{obj}} + w_2 m_j + w_3 P_{j,\mathrm{motion}}\right), \qquad \pi_j \in (0,1). \tag{N9}$$

$$A_{ij} = \frac{Q_i K_j^\top}{\sqrt{d}} + \log(\varepsilon + \pi_j), \qquad \alpha_{ij} = \mathrm{softmax}_j(A_{ij}). \tag{N10}$$

A log-prior multiplicatively reweights attention after softmax. This is a plausible design, not a Bayesian guarantee or a proven improvement. Motion can be confounded by camera ego-motion; drop it if reliable previous-frame information is absent.

7 Objectness is not uncertainty

$$\mathcal{L}_{\mathrm{obj}} = \mathrm{FocalLoss}(\widehat{H}, H_{\mathrm{center}}). \tag{N11, optional}$$

The existing centre heatmap estimates object-centre probability (objectness), not uncertainty. It may be retained as an auxiliary dense loss or for top-k query initialisation.

$$\mathcal{L}_{\text{box-NLL}} = \sum_k \left[|b_k - \widehat{b}_k| \exp(-s_k) + s_k \right], \qquad s_k = \log(\text{scale}_k). \tag{N12, optional}$$

This optional Laplace negative log-likelihood models aleatoric localisation noise. It is distinct from memory anomaly evidence and epistemic uncertainty, and should be deferred until the detector is stable and validation data support calibration assessment.

8 Total objective and gradient paths

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{pair-det}} + \lambda_{\text{inv}}\mathcal{L}_{\text{inv}} + \lambda_{\text{obj}}\mathcal{L}_{\text{obj}}. \tag{N13}$$

For the smallest POC, use $\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{pair-det}}$. Add $\mathcal{L}_{\text{inv}}$ and $\mathcal{L}_{\text{obj}}$ only as controlled ablations.

Term	**Decision**	**Reason / gradient path**
$\mathcal{L}_{\text{det}}$ on x and x^+	Required	Trains projection/encoder, decoder and heads; not backbone or banks.
$\mathcal{L}_{\text{mem}} = \text{mean}(E)$	Remove	No trainable path under frozen backbone and fixed memory.
$\mathcal{L}_{\text{rob}} \to \exp(-E)$	Remove	Target is a deterministic function of input evidence.
$\mathcal{L}_{\text{gate}}$ / preserve	Remove	Anomaly-dependent gate can suppress degraded tokens; preserve term only patches the conflict.
$\mathcal{L}_{\text{inv}}$	Optional	Trains downstream representations; detection loss must anchor them.
$\mathcal{L}_{\text{obj}}$	Optional	Supervises objectness, not uncertainty.
Box NLL	Defer	Requires calibration evaluation.

9 Evidence versus hypotheses

Conclusion	**Status**
$\mathcal{L}_{\text{mem}}$ has no gradient path with frozen backbone and fixed bank.	Supported by formulation/code review.
$\exp(-E)$ target suppresses high-energy tokens.	Supported by formulation.
Current heatmap is objectness, not uncertainty.	Supported by loss definition.
High E identifies damaged TCDs.	Unsupported by current measurements.
Two-bank margin separates damaged TCDs from background.	Plausible; needs validation.
Log-space prior improves attention.	Plausible; needs validation.
Stride-16 is too coarse for these boxes.	Plausible from box statistics; not tested.
Failed tiny-set overfit invalidates the concept.	Unsupported; signals implementation/training concerns, not a verdict.

10 Minimal proof-of-concept plan

10.1 Stage A — no new detector training

- Verify object/background membership in the memory banks visually.
- Visualise d_{obj}, d_{bg} and $m = d_{\text{bg}} - d_{\text{obj}}$ on selected clean and mud-degraded TCD examples.
- Manually tag approximately 30–50 damaged/occluded examples if available; report qualitative results and failure cases.
- Measure token separability for TCD versus background, then separately inspect clean-versus-damaged behaviour.
- Optionally re-score a baseline detector with a box-level summary of evidence; compare damaged-subset recall at a fixed false-positive rate.

10.2 Stage B — smallest trained PatchDETR

- Frozen stride-8 features, two memory banks, evidence-conditioned tokens and a small DETR decoder.
- Train with paired detection loss first; add invariance only as an ablation.
- Compare baseline DETR → evidence tokens → optional attention prior.
- Before long runs, verify actual gradient paths and run a small sanity check. Do not interpret this as a performance claim.

Current data cannot establish robust generalisation across sites/weather, calibrated uncertainty, or statistically reliable superiority on damaged TCDs. With approximately 800 train images, 300 test images, 500 memory images and no comprehensive damage labels, the credible outcome is an exploratory POC and a rationale for larger-scale Bosch validation.

11 Mapping to the existing code

Equation	Current code area from review	Change
N1	`model.py: ResNetBackbone`	Keep frozen; use stride-8 or a downstream neck.
N2–N4	`model.py` retrieval; `memory.py` bank build	Split object/background banks; expose evidence vector.
N5	`PatchCvTModule.forward`	Feed evidence tokens; remove R/S gate.
N6–N7	DETR matcher and losses	Keep standard set loss; apply to clean and degraded images.
N8	`losses.py` semantic term	Use aligned tokens and stop-gradient clean target.
N9–N10	Attention bias / decoder layer	Optional log-prior replacement for current additive bias.
N11	Heatmap head/loss	Rename/document as objectness.
N12	Box head and loss	Defer predicted variance.
N13	Loss composition	Remove $\mathcal{L}_{\text{mem}}$, $\mathcal{L}_{\text{rob}}$, $\mathcal{L}_{\text{gate}}$ and preserve term.

12 Deviations from the original PatchCvT formulation

The original PatchCvT formulation should remain intact as a reference. The proposed PatchDETR integration differs deliberately:

- Replace the mixed bank with separate object and background banks.
- Remove the R/S gate from the minimum formulation because its current target can suppress high-energy/degraded tokens.
- Remove $\mathcal{L}_{\text{mem}}$ under the frozen-backbone/fixed-bank constraint because it has no trainable gradient path.
- Rename the current heatmap as objectness. A distinct uncertainty estimate is optional and deferred.
- Retain clean/degraded invariance only as an optional downstream loss with a detached clean target.

These changes do not claim that original PatchCvT is generally invalid; they state that some terms are not useful by default under the current frozen-backbone constraint and targets.

13 Recommendation

Do not launch another long training run yet. First test whether the two-bank evidence maps carry interpretable signal on selected examples. If they fail to separate TCD regions from background, or show no meaningful response to degradation, a larger PatchDETR training run is unlikely to rescue the central memory-evidence hypothesis without further conceptual changes.

Defensible current claim: PatchDETR proposes to condition DETR-style detection on frozen-feature memory evidence to preserve useful object cues under visual degradation. Its value for damaged TCDs remains to be validated.